

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 324 — Past Questions & Materials
by Renance

Course: MAR 324
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200, 300

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Paper 1 of 4 — MAR 324 - 2024/2025 - Second Semester - Marine Engineering Materials

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
 DEPARTMENT OF MARINE ENGINEERING
 SECOND SEMESTER EXAMINATION 2024/2025
 COURSE CODE & TITLE: MARINE ENGINEERING MATERIALS, MAR324
 DATE: 23/09/2025, TIME: 2HRS, 45 MINS INSTRUCTION
 • READ CAREFULLY BEFORE ANSWERING ANY QUESTION.
 • ATTEMPT QUESTION ONE AND ANY OTHER THREE (3) QUESTIONS OF YOUR CHOICE

QUESTION ONE (Compulsory)

- Estimate the activation energy for creep for the S-590 alloy having the steady state creep behaviour shown in Figure A. Use the data taken at stress level of 300 Mpa (43,500psi) and temperature of 650°C and 730°C. Assume that the stress component is independent of temperature [10.5 marks] (Take $R = 8.314 \text{ J/mol} \cdot \text{K}$)
- Estimate the creep strain rate at 600°C and 300 Mpa [7 marks]

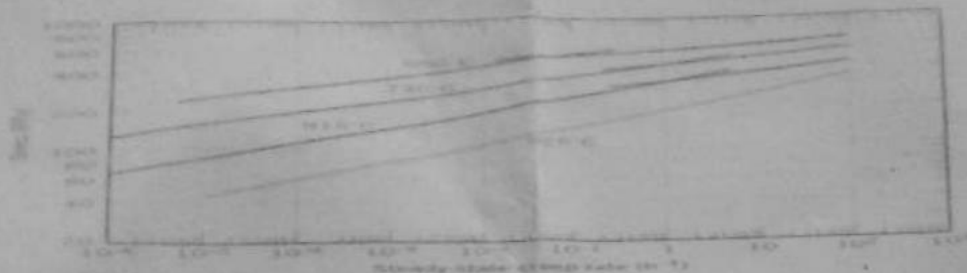


Fig: A: Creep strain rate

QUESTION TWO

A newly built ocean-going tugboat is required by the MLS shipping company for deploying bulk cargo and passengers in coastal waters with emphasis on speed and strength to weight ratio. The proposed choice of materials for the hull are mild steel, Gray cast iron, and aluminum. A performance selection is to be carried out to choose the best. Various experts have been consulted, and assuming your professional inputs are required:

- Describe the coastal water environment with respect to density, and hydrodynamic forces [4 marks]
- Evaluate the possibility of producing the ocean tugboat using a flowchart that systematically and technically explains how it can be achieved from material selection to the production stage using the materials cited [7.5 marks]
- If ocean tugboat is considered to be a beam fixed at the port end, estimate the deflection at the free starboard end when a load is applied on the materials and ascertain which material has less deflection using Table Y [3marks]
- Use the stress-strain relationship to demonstrate the performance of the materials with respect to yield strength and proffer which material is suitable for application in the said environment with respect to strength to weight ratio [3 marks]

QUESTION THREE

With reference to question two

- Estimate the mean time to failure for each of the ocean tugboat materials if the tugboat fails the submersibility test using the operating data in Table Y. [3.5 mark]
- Applying the data in table Y, determine the best reliability state for the ocean-going tugboat materials if it is to be operated at 1000, 1200, 1400, 1600, and 1800 hrs., using exponential and two parameter Weibull distribution [5 marks]
- Plot a graph of reliability against time using your results in part (b) [2.5 marks]
- Using the cost quotation in Table Y for the three ocean-going tugboat materials, estimate the optimal life cycle cost analysis to adopt (direct cost estimation) [6.5 marks]

QUESTION FOUR

Utilizing the information from question two:

- Numerically establish the stability factor for the box-shape ocean-going tugboat and use the operational stability data to predict the suitable material for the ocean-going tugboat that would be stable [6 marks]
- Based on your assessment of the material to be used for the tugboat boat design with respect to the environment and company needs, assess the level of interaction between the material and the environmental and consider loading operations on the deck of the tugboat [7 marks]

- c) Discuss the findings on your choice of material in part (b) for the design and production of the ocean-going tugboat and evaluate the overall results with emphasis on LCCA, Reliability (R), weight to strength ratio, mean time to failure and stability [6 marks]
- d) Suggest a maintenance concept to be adopted by the company to ensure that the required choice of material for the ocean-going tugboat design meets its expected useful life [1 mark]

QUESTION FIVE

During the load test analysis on the tugboat main deck, stress was analyzed to examine which of the materials can perform better before failure. Using the data in question 2 and Table Y, hence:

- Estimate the stress intensity factor for each material assuming the deck of the ocean-going tugboat is an infinitely large sheet subjected to a gross stress of 450 Mpa resulting in a central crack 5/m-cm [6 marks]
- Determine the plastic zone size at the crack tip of each material [3 marks]
- How valid is the plastic zone corrector factor for each material? [3 marks]
- Looking at your numerical findings and the needs of the company, what would be your judgement? discuss the likely implications or maintenance measures you will advise the company to observe [5.5 marks]

QUESTION SIX

The performance selection of a suitable material or materials for marine vessels design and the environment where they will be deployed is essential. Hence:

- Discuss five mechanical properties that should guide the mind of a material engineer during material selection phase [5 marks]
- Suggest a model that can be used to estimate the reliability of a marine equipment over the entire life using a suitable diagram [4 marks]
- Differentiate creeping from fracture using any metallic material as a case study [4.5 marks]
- Discuss how you can apply the knowledge of the course " Marine engineering materials" in the maritime and offshore sector? [4marks]

Table Y: Operational data

Materials	Mild steel	Gray Cast iron	Aluminum
Numbers of failures	8	7	6
Operating hours	550, 480, 680, 790, 860 and 620	550, 480, 680, 790, 860 and 620	550, 480, 680, 790, 860 and 620
Scale parameter (η)	800	800	800
Shape parameter (β)	0.6	0.6	0.6
Procurement cost	£290,000	£300,000	£270,000
Expected useful life in yrs	10	10	10
Annual failure rate	0.06	0.08	0.07
Cost of a failure	£3000	£2,000	£2500
Annual interest rate	6%	6%	6%
Annual operating cost	£65000	£6,000	£7000
Length (L) of the beam	$1.5 \times 10^3 \text{ mm}$	$1.8 \times 10^3 \text{ mm}$	$1.6 \times 10^3 \text{ mm}$
Second Moment of Inertia (I)	$33.75 \times 10^6 \text{ mm}^4$	$33.75 \times 10^6 \text{ mm}^4$	$33.75 \times 10^6 \text{ mm}^4$
E	200 GPa	200 GPa	200 GPa
Force applied at the free end of the beam	20 KN	20 KN	20 KN
Yield strength	500 Mpa	207 Mpa	460 Mpa

STABILITY DATA

Width of Tugboat (b)	1.0 m	1.2 m	1.4m
Draft (d)	0.48 m	0.48m	0.48m
KG	5.0 m	5.0 m	5.0 m
Tensile strength	400 Mpa	250 Mpa	210 Mpa

Paper 2 of 4 — MAR 324 Non-Ferrous Metals and Alloys

Level 300 • Second Semester • Handout

Non-Ferrous Metals and Alloys

1. Aluminum

Aluminum ore is found as a hydrated aluminum oxide, called bauxite. The impurities present in it are oxides of iron, silicon, and titanium. The process therefore, is to separate aluminum oxide from these impurities. For this purpose, bauxite is fused in electric furnace and carbon is added to reduce the impurities, which form sludge and can be removed. As a result of this refining pure aluminum oxide is separated from the impurities. Then an electrolytic bath is used to reduce aluminum from its oxide. As the electrolytic process proceeds the oxygen escapes through the bath and molten aluminum collects at the bottom, from where it is periodically tapped off.

Properties:

1. High electrical conductivity
2. High heat conductivity
3. Good resistance to corrosion
4. Highly ductile
5. Light in weight

Application

1. Used for heavy conductor and bus bar work.
2. In domestic utensils and other heat conductive appliances.
3. Used in manufacture of containers for chemical industries.
4. It can be easily worked, extruded, rolled, drawn, and forged.

2. Copper

It is not available in pure form under the earth. It is extracted from its ores through a series of processes. Copper pyrites are the main ores used for extracting copper. The copper ore is first roasted to drive out the water, CO₂ and sulphur. It is followed by melting in a reverberatory furnace of the type used for wrought iron. Silica is added to the charge to form slag with impurities like iron and alumina, etc. The molten metal is tapped and transferred to a convertor where air is blown through it to burn the impurities. This results in the production of crude form of copper, known as blister copper, containing 68% purity. Final refining is done by an

1

electrolytic process, pure copper depositing on the cathode. This gives a highly pure (99.9%) copper which is re melted and cast into suitable shapes.

Properties:

1. High electrical conductivity.
2. High heat conductivity
3. Good corrosion resistance
4. High ductility
5. Light in weight

Application:

1. Used as electrical conductor in various shapes and forms viz., sheet and contacts.
2. Used in heat exchanger and heating vessel and appliances.
3. Used for providing base coating on steel prior to nickel and chromium plating.
4. Can be easily cold worked, rolled, and drawn.
5. Used in various appliances where light weight is desired.

3. Magnesium

Principle sources for obtaining magnesium are natural salt brines, sea water, and water liquors obtained from potash industry and ores. The principle ores are magnesite, dolomite and carnallite. Various processes have been developed for its extraction, but the most popular and widely used one is electrolytic process

Properties

1. Lightest of all metals, weighing about two-third of aluminum.
2. It may be sand, gravity and pressure die-cast.
3. Its casting is pressure tight and obtains good surface finish.
4. It may be easily formed, drawn, forged and machined with high accuracy.
5. Addition of 10% aluminum and small amount of zinc and manganese improve its strength and casting characteristic.
6. Addition of 2% Mn helps in its easy forming into plates and sheets and extrusion works.

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Application: Motor car gear box, differential housing, and portable tools

4. Zinc

The zinc ore is first concentrated through suitable process. This concentrate is fed into a retort with suitable amount of carbonaceous material (say coal). Several retorts are housed in one furnace and their temperature raised to 1100°C. Zinc emerges as a vapour and is passed through a condenser. By rapid cooling the zinc vapour may be quickly converted into powdered zinc.

Properties

1. High corrosion resistance
2. Low melting point
3. High fluidity

Application: Widely used as a protective coating on iron and steel. Low melting point and high fluidity make it the most suitable metal for pressure die-casting.

5. Lead

Lead ores are generally found as oxides or sulphides. Other impurities present in the ores are iron, copper and zinc etc. The prepared ore concentrate, together with the flux (lime and silica), is fed into a small blast furnace, where the temperature is raised to about 1010°C. The lead is melted and a liquid slag formed of the impurities. Both slag and molten lead are tapped at intervals. Further refining is carried out in a reverberatory furnace where an oxidizing atmosphere is maintained to burn out the impurities.

Properties:

1. Good corrosion resistance
2. Good resistance to chemical action.
3. It is soft heavy and malleable
4. It is alloyed with brass and steel to impart them free cutting properties.

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Application

1. Used for water pipe and roof protection
2. Used for acid bath and containers in chemical industry.
3. Can be easily worked and shaped.
4. Used as an alloying element in making soft solders and plumber's solders.

6. Tin

The most prominent tin ore is cassiterites. It also carried compounds of copper, iron, lead antimony, bismuth and zinc etc. As usual an ore concentrate is prepared. This concentrate is roasted to drive off excess arsenic and sulphur. The roasted ore is transferred to a reverberatory furnace, where it is heated. Anthracite is added to the charge which reacts chemically to separate tin, the latter sinking to the bottom of the furnace. From there it is tapped at intervals. This crude tin is re melted and re refined further.

Properties

1. Good resistance to acid corrosion
2. It is soft, has good plasticity and can be easily worked.
3. It can be easily rolled in thin foils, but cannot be drawn due to low strength..

Application

It is used as an alloying element in soft solders, bronzes bearing metals.

7. Nickel

Its extraction process consists of first roasting the ore, followed by melting in a small blast furnace. Lime stone and quartz are added as flux. They form slag with impurities. Coke is used as fuel. Crude molten nickel tapped off periodically from the bottom of the furnace. This crude metal is further refined in a Bessemer converter followed by treating with sulphuric acid to extract pure nickel. Copper is separated as a copper sulphate.

Properties

1. It has good resistance to both acid and alkali corrosion

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2. It has high tensile strength and can be easily worked cold and hot.

Application

1. It is widely used in food processing.
2. It is plated on steel to provide a corrosion resistant surface
3. It is important alloying element with steel.

Non-ferrous alloys

Due to poor physical and mechanical properties and high costs the non-ferrous metals are seldom

used in their in pure state. But their alloys carry very good physical and mechanical properties and are widely used. In general they have lower strength, hardness, and modulus of elasticity than irons and steels. However they carry their own importance due some exceptional properties possess, like lightness, ease in fabrication, good machinability, high resistance to corrosion, attractive appearance and good cast ability etc. Some commonly used non-ferrous alloys are described in the following articles:

1. Brasses

All brasses are basically alloys of copper and zinc. There are two main varieties of brasses:

1. Alpha brass (Zn up to 37%) - for cold working.
2. Alpha beta brass (Zn 33% to 46%) - for hot working.

Alpha brasses are very ductile and can be readily cold worked without any chances of fracture. They can be cold rolled into sheet, drawn into wire, deep drawn and drawn into tubes. In these brasses, as the proportion of zinc increases, their strength increases but ductility decreases. They are worked hardened when subjected to intensive cold working, but ductility can be regained by annealing them at 6000 C. Slow cooling provides maximum ductility, but for common uses they may be water quenched. Deep drawing of this brass requires periodical annealing during the process.

Alpha beta brasses lose strength at high temperature but become very plastic. It, therefore respond very well to hot rolling, hot extrusion, hot stamping and casting, etc. when cold worked, fracture are likely to develop.

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Common types of brasses in engineering use are the following:

- a) Cartridge brass: - It has 70% Cu and 30% Zn. It is very strong and ductile. It is used for wide range of drawn components like cartridge cases, head lamp reflector, radiator shells and drawn tubes.
- b) Muntz metals: - It contains 60% Cu and 40% Zn and can be cast, rolled, extruded and stamped. It is a sort of general purpose alloy having good resistance to corrosion. It is used for casting pump parts, valve, taps and other similar items.
- c) Naval brass: - It contains 60% Cu, 39% Zn and 1% tin. It is more or less similar in composition to Muntz metals except that 1% Zn is replace by 1% tin. As a result of this change the resistance to sea water corrosion is vastly improved. This alloy is therefore, widely used for cast and forged fittings for ships.
- d) Admiralty brass: - It contains 70% Cu, 29% Zn and 1% tin. It is similar to cartridge brass in composition except that 1% Zn is replaced by 1% tin. It can be cold worked and has good resistance to sea water corrosion. It is cold drawn into tubes and rolled into sheets and bars. It is widely used in ship fitting, bolts, nuts, washer and other items subjected to sea water corrosion.
- e) Gilding brass: - it contain up to 15% zinc and the rest Cu. It is very cold working alloy and is used for jewelry, decorative and ornamental work. It is commercially available as cold rolled strip, wire or sheet. Its color according to the percentage of Zn varies from red

to bright yellow. It is also called Gilding metal.

f) Delta brass: - It is also known as delta metal, it consists of 60% of Cu, 37% Zn and 3% iron. It can be easily hot worked, forged, rolled, extruded and cast. It has fairly good tensile strength after hot working and casting. It also has a good corrosion resistance. It can suitably replace steel castings.

g) Free cutting brass: - It contains 60% Cu, 37%Zn and 3% Pb. It is specially used in machining work, such as producing components from bar stock on turret and automatic lathe. It is also used for making cast, forged, or stamped, blanks to be used for further machining. With this metal very high speeds and feeds can be employed in machining.

h) Beta brass: - It contains 50% Cu, and 50% Zn. Higher percentage of zinc render it hard and brittle, but it softens quickly when heated and melts at 8700 C. Its main application is brazing solder.

i) Silicon brass: - It contains 80% Cu, 16% Zn and 4% Si. It responds well to welding and is widely used for refrigerators and fire extinguisher shells. It can also be easily sand or

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gravity die cast, hot stamped and extruded. It can be used as a cheaper substitute for phosphor bronze.

2. Bronzes

Bronze is basically an alloy of copper and tin. In general, it possesses superior mechanical properties and corrosion resistance than brass. Those containing up to 8% tin are called working bronze. They can be easily cold worked, rolled, formed and drawn. They are available in various forms, as strip, wire and sheet etc.

With the increase in tin content, its strength and corrosion resistance increase. It is then known as hot working bronzes. Small addition of phosphorous further improves its strength, ductility and bearing properties. The amount of phosphorous added is 0.5%. This is then known as phosphorous bronze.

a. Phosphorous bronze: - Widely used in the several manufacturing units, we present an array of phosphor bronze alloys rods, phosphor bronze bars, phosphor bronze squares and many more. These phosphor bronze alloys offer resistance towards fatigue and chemical corrosion, so they are used for varied application in various industries such as in heavy engineering, defense, aviation, electrical and more.

b. Gun metal: - The term 'gunmetal' is applied to a family of copper-based casting alloys containing between 2-11% tin and 1-10% zinc. Modified forms may contain, in addition, such elements as lead (up to 7%) and nickel (up to 6%) when the alloys are classified as 'leaded gunmetal' and 'nickel gunmetal'. Gunmetal is noted for the manufacture of intricate castings required to be pressure tight such as valves, pipe fittings and pumps.

c. Aluminum bronze: - Aluminum bronzes are used for their combination of high strength, excellent corrosion and wear resistance. Aluminum bronze alloys typically contain 9-12% aluminum and up to 6% iron and nickel. Alloys in these composition limits are hardened by a combination of solid solution strengthening, cold work, and precipitation of an iron rich

phase. High aluminum alloys are quenched and tempered. Aluminum bronzes are used in marine hardware, shafts and pump and valve components for handling seawater, sour mine waters, and industrial process fluids. They are also used in applications such as heavy duty sleeve bearings, and machine tool ways.

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d. Silicon bronze: - it contains 1 to 4% Si, 0.25% to 1.25% Mn, 0.5% to 1% iron (if added) and the rest copper. Small addition of upto 0.5% Pb will improve machinability. It has high strength, toughness, and corrosion resistance. It can be readily hot worked. With low silicon content it can be safely cold worked also. It is widely used for boiler parts, tanks marine hardware and similar other item.

e. Manganese bronze: - It contains 55% to 60% copper, 38% to 42% Zn, upto 1.5% tin, upto 2% iron,

upto 1.5% Al and upto 3.5% Mn. It has superior mechanical properties and high corrosion resistance. It has poor respond to cold working but can be readily hot worked. It is used for such part

where high strength and corrosion resistance are desired, such as in ship propellers and rudders, etc.

Aluminum alloys

a. Duralumin: - It contains 4% Cu, 0.5% Mg, 0.5% Mn and the rest aluminum. Duralumin alloys are relatively soft, ductile, and workable in the normal state; they may be rolled, forged, extruded, or drawn into a variety of shapes and products. Their light weight and consequent high strength per unit weight compared to steel suit them for aircraft construction. Because duralumin loses strength in welding, a special laminated sheet form called alclad is used for aircraft construction; it has thin surface layers of pure aluminum or a corrosion-resistant aluminum alloy covering the strong duralumin core.

b. Aluminum casting alloy: - It contains 90% Al, 8% Cu, 1% iron and 1% Si. It has good strength, hardness and machinability. It may be sand, pressure or gravity die cast.

c. Y-alloy: - It contains 93% Al, 4% Cu, 2% Ni, and 1% Mg. Its principle use is as a casting alloy. It maintains its strength at elevated temperatures, and is used for pistons of I.C. engines. A treatment of Y-alloy castings, consisting of quenching in boiling water from a temperature of 5100 C and then aging for 5 days, develops very good mechanical properties in them. It is also used in strip and sheet forms.

Nickel alloys

a. German silver: - The name German silver or nickel silver is misleading since it has no silver content, but is an alloy of 60% Cu, 20% Ni and 20% Zn. It resembles silver but instead of the soft white color it has a yellow-grey tinge.

German silver is used extensively in the hollow-ware and flatware industries as E.P.N.S. (electro-plated nickel silver), which is a nickel silver item with a plating of silver. Nickel silver is slightly more difficult to work for jewelry purpose, than copper and other metals, since it is a harder alloy. Again it is prone to tarnish on exposure to air and will turn the skin green with prolonged contact.

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- b. Constantan: - It contains 45% Ni, 55% Cu. It has high specific resistance, which is unaffected by temperature variation. It is used for accurate resistors, thermocouples, Wheatstone bridge, low temperature heater and resistances.
- c. Monel metal: - It contains 68% Ni, 30% Cu, 1% iron and small additions of Mn and other elements. It is a nickel-copper alloy with high strength and excellent corrosion resistance in a range of media including sea water, hydrofluoric acid, sulfuric acid, and alkalis. Used for marine engineering, chemical and hydrocarbon processing equipment, valves, pumps, shafts, fittings, fasteners, and heat exchangers.
- d. Inconel: - It contains 80% Ni, 14% Cr and 6% iron. Inconel Alloys Are Generally Known For Their Resistance To Oxidation And Their Ability To Maintain Their Structural Integrity In High Temperature Atmospheres. There Are Several Inconel Alloys That Are Used In Applications That Require A Material That Does Not Easily Succumb To Caustic Corrosion, Corrosion Caused Purity Water, And Stress-Corrosion Cracking. While Each Variation Of Inconel Has Unique Traits that make it effective in different circumstances, the majority of the alloys are used frequently in the chemical industry.
- e. Nichrome: - Like Inconel it is also a nickel chromium alloy which is extensively used in electrical appliances as resistance wire.
- f. Incoloy: - It is also a Ni based alloy which is widely used as a high temperature alloy. It consists of 42% Ni, 13% Cr, 6% Mo, 2.4% Ti, 0.04% C and rest iron.
- g. K-Monel: - It possesses similar composition as monel, but about 3 to 4% Al is added to it. It carries similar application as monel, but has better mechanical properties than that.
- h. Nimonic alloy: - It contains 80% Ni and 20% Cr. It has high strength and ability to operate under intermittent heating and cooling conditions. It is widely used in gas turbine engines.

Bearing metals

Bearing metal should possess the following important characteristics:

1. It should have enough compressive strength to possess adequate load carrying capacity.
2. It should have good plasticity to allow for small variation in alignment and fitting.
3. It should have low co-efficient of friction to avoid excessive heating.

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Some important bearing metals are the following:

- a. Babbitt metal: - It is white metal containing 85% tin, 10% Sb and 5% Cu. It is used for heavy duty vehicles.
- b. Lead alloy: - It contains 40% lead and 60% Cu. It may be cast in position or fused as a thin shell to a bronze or steel reinforcing shell outside.
- c. Phosphor bronze: - With 10% tin is used for light load low speed bearings. It can be sand and centrifugal cast.

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Paper 3 of 4 — MAR 324 Non-Ferrous Metals and Alloys

Level 300 • Second Semester • Handout

Non-Ferrous Metals and Alloys

1. Aluminum

Aluminum ore is found as a hydrated aluminum oxide, called bauxite. The impurities present in it are oxides of iron, silicon, and titanium. The process therefore, is to separate aluminum oxide from these impurities. For this purpose, bauxite is fused in electric furnace and carbon is added to reduce the impurities, which form sludge and can be removed. As a result of this refining pure aluminum oxide is separated from the impurities. Then an electrolytic bath is used to reduce aluminum from its oxide. As the electrolytic process proceeds the oxygen escapes through the bath and molten aluminum collects at the bottom, from where it is periodically tapped off.

Properties:

1. High electrical conductivity
2. High heat conductivity
3. Good resistance to corrosion
4. Highly ductile
5. Light in weight

Application

1. Used for heavy conductor and bus bar work.
2. In domestic utensils and other heat conductive appliances.
3. Used in manufacture of containers for chemical industries.
4. It can be easily worked, extruded, rolled, drawn, and forged.

2. Copper

It is not available in pure form under the earth. It is extracted from its ores through a series of processes. Copper pyrites are the main ores used for extracting copper. The copper ore is first roasted to drive out the water, CO₂ and sulphur. It is followed by melting in a reverberatory furnace of the type used for wrought iron. Silica is added to the charge to form slag with impurities like iron and alumina, etc. The molten metal is tapped and transferred to a convertor where air is blown through it to burn the impurities. This results in the production of crude form of copper, known as blister copper, containing 68% purity. Final refining is done by an

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electrolytic process, pure copper depositing on the cathode. This gives a highly pure (99.9%) copper which is re melted and cast into suitable shapes.

Properties:

1. High electrical conductivity.
2. High heat conductivity
3. Good corrosion resistance
4. High ductility
5. Light in weight

Application:

1. Used as electrical conductor in various shapes and forms viz., sheet and contacts.
2. Used in heat exchanger and heating vessel and appliances.
3. Used for providing base coating on steel prior to nickel and chromium plating.
4. Can be easily cold worked, rolled, and drawn.
5. Used in various appliances where light weight is desired.

3. Magnesium

Principle sources for obtaining magnesium are natural salt brines, sea water, and water liquors obtained from potash industry and ores. The principle ores are magnesite, dolomite and carnallite. Various processes have been developed for its extraction, but the most popular and widely used one is electrolytic process

Properties

1. Lightest of all metals, weighing about two-third of aluminum.
2. It may be sand, gravity and pressure die-cast.
3. Its casting is pressure tight and obtains good surface finish.
4. It may be easily formed, drawn, forged and machined with high accuracy.
5. Addition of 10% aluminum and small amount of zinc and manganese improve its strength and casting characteristic.
6. Addition of 2% Mn helps in its easy forming into plates and sheets and extrusion works.

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Application: Motor car gear box, differential housing, and portable tools

4. Zinc

The zinc ore is first concentrated through suitable process. This concentrate is fed into a retort with suitable amount of carbonaceous material (say coal). Several retorts are housed in one furnace and their temperature raised to 1100°C. Zinc emerges as a vapour and is passed through a condenser. By rapid cooling the zinc vapour may be quickly converted into powdered zinc.

Properties

1. High corrosion resistance
2. Low melting point
3. High fluidity

Application: Widely used as a protective coating on iron and steel. Low melting point and high fluidity make it the most suitable metal for pressure die-casting.

5. Lead

Lead ores are generally found as oxides or sulphides. Other impurities present in the ores are iron, copper and zinc etc. The prepared ore concentrate, together with the flux (lime and silica), is fed into a small blast furnace, where the temperature is raised to about 1010°C. The lead is melted and a liquid slag formed of the impurities. Both slag and molten lead are tapped at intervals. Further refining is carried out in a reverberatory furnace where an oxidizing atmosphere is maintained to burn out the impurities.

Properties:

1. Good corrosion resistance
2. Good resistance to chemical action.
3. It is soft heavy and malleable
4. It is alloyed with brass and steel to impart them free cutting properties.

3

Application

1. Used for water pipe and roof protection
2. Used for acid bath and containers in chemical industry.
3. Can be easily worked and shaped.
4. Used as an alloying element in making soft solders and plumber's solders.

6. Tin

The most prominent tin ore is cassiterites. It also carried compounds of copper, iron, lead antimony, bismuth and zinc etc. As usual an ore concentrate is prepared. This concentrate is roasted to drive off excess arsenic and sulphur. The roasted ore is transferred to a reverberatory furnace, where it is heated. Anthracite is added to the charge which reacts chemically to separate tin, the latter sinking to the bottom of the furnace. From there it is tapped at intervals. This crude tin is re melted and re refined further.

Properties

1. Good resistance to acid corrosion
2. It is soft, has good plasticity and can be easily worked.
3. It can be easily rolled in thin foils, but cannot be drawn due to low strength..

Application

It is used as an alloying element in soft solders, bronzes bearing metals.

7. Nickel

Its extraction process consists of first roasting the ore, followed by melting in a small blast furnace. Lime stone and quartz are added as flux. They form slag with impurities. Coke is used as fuel. Crude molten nickel tapped off periodically from the bottom of the furnace. This crude metal is further refined in a Bessemer converter followed by treating with sulphuric acid to extract pure nickel. Copper is separated as a copper sulphate.

Properties

1. It has good resistance to both acid and alkali corrosion

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2. It has high tensile strength and can be easily worked cold and hot.

Application

1. It is widely used in food processing.
2. It is plated on steel to provide a corrosion resistant surface
3. It is important alloying element with steel.

Non-ferrous alloys

Due to poor physical and mechanical properties and high costs the non-ferrous metals are seldom

used in their in pure state. But their alloys carry very good physical and mechanical properties and are widely used. In general they have lower strength, hardness, and modulus of elasticity than irons and steels. However they carry their own importance due some exceptional properties possess, like lightness, ease in fabrication, good machinability, high resistance to corrosion, attractive appearance and good cast ability etc. Some commonly used non-ferrous alloys are described in the following articles:

1. Brasses

All brasses are basically alloys of copper and zinc. There are two main varieties of brasses:

1. Alpha brass (Zn up to 37%) - for cold working.
2. Alpha beta brass (Zn 33% to 46%) – for hot working.

Alpha brasses are very ductile and can be readily cold worked without any chances of fracture. They can be cold rolled into sheet, drawn into wire, deep drawn and drawn into tubes. In these brasses, as the proportion of zinc increases, their strength increases but ductility decreases. They are worked hardened when subjected to intensive cold working, but ductility can be regained by annealing them at 6000 C. Slow cooling provides maximum ductility, but for common uses they may be water quenched. Deep drawing of this brass requires periodical annealing during the process.

Alpha beta brasses lose strength at high temperature but become very plastic. It, therefore respond very well to hot rolling, hot extrusion, hot stamping and casting, etc. when cold worked, fracture are likely to develop.

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Common types of brasses in engineering use are the following:

- a) Cartridge brass: - It has 70% Cu and 30% Zn. It is very strong and ductile. It is used for wide range of drawn components like cartridge cases, head lamp reflector, radiator shells and drawn tubes.
- b) Muntz metals: - It contains 60% Cu and 40% Zn and can be cast, rolled, extruded and stamped. It is a sort of general purpose alloy having good resistance to corrosion. It is used for casting pump parts, valve, taps and other similar items.
- c) Naval brass: - It contains 60% Cu, 39% Zn and 1% tin. It is more or less similar in composition to Muntz metals except that 1% Zn is replace by 1% tin. As a result of this change the resistance to sea water corrosion is vastly improved. This alloy is therefore, widely used for cast and forged fittings for ships.
- d) Admiralty brass: - It contains 70% Cu, 29% Zn and 1% tin. It is similar to cartridge brass in composition except that 1% Zn is replaced by 1% tin. It can be cold worked and has good resistance to sea water corrosion. It is cold drawn into tubes and rolled into sheets and bars. It is widely used in ship fitting, bolts, nuts, washer and other items subjected to sea water corrosion.
- e) Gilding brass: - it contain up to 15% zinc and the rest Cu. It is very cold working alloy and is used for jewelry, decorative and ornamental work. It is commercially available as cold rolled strip, wire or sheet. Its color according to the percentage of Zn varies from red

to bright yellow. It is also called Gilding metal.

f) Delta brass: - It is also known as delta metal, it consists of 60% of Cu, 37% Zn and 3% iron. It can be easily hot worked, forged, rolled, extruded and cast. It has fairly good tensile strength after hot working and casting. It also has a good corrosion resistance. It can suitably replace steel castings.

g) Free cutting brass: - It contains 60% Cu, 37%Zn and 3% Pb. It is specially used in machining work, such as producing components from bar stock on turret and automatic lathe. It is also used for making cast, forged, or stamped, blanks to be used for further machining. With this metal very high speeds and feeds can be employed in machining.

h) Beta brass: - It contains 50% Cu, and 50% Zn. Higher percentage of zinc render it hard and brittle, but it softens quickly when heated and melts at 8700 C. Its main application is brazing solder.

i) Silicon brass: - It contains 80% Cu, 16% Zn and 4% Si. It responds well to welding and is widely used for refrigerators and fire extinguisher shells. It can also be easily sand or

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gravity die cast, hot stamped and extruded. It can be used as a cheaper substitute for phosphor bronze.

2. Bronzes

Bronze is basically an alloy of copper and tin. In general, it possesses superior mechanical properties and corrosion resistance than brass. Those containing up to 8% tin are called working bronze. They can be easily cold worked, rolled, formed and drawn. They are available in various forms, as strip, wire and sheet etc.

With the increase in tin content, its strength and corrosion resistance increase. It is then known as hot working bronzes. Small addition of phosphorous further improves its strength, ductility and bearing properties. The amount of phosphorous added is 0.5%. This is then known as phosphorous bronze.

a. Phosphorous bronze: - Widely used in the several manufacturing units, we present an array of phosphor bronze alloys rods, phosphor bronze bars, phosphor bronze squares and many more. These phosphor bronze alloys offer resistance towards fatigue and chemical corrosion, so they are used for varied application in various industries such as in heavy engineering, defense, aviation, electrical and more.

b. Gun metal: - The term 'gunmetal' is applied to a family of copper-based casting alloys containing between 2-11% tin and 1-10% zinc. Modified forms may contain, in addition, such elements as lead (up to 7%) and nickel (up to 6%) when the alloys are classified as 'leaded gunmetal' and 'nickel gunmetal'. Gunmetal is noted for the manufacture of intricate castings required to be pressure tight such as valves, pipe fittings and pumps.

c. Aluminum bronze: - Aluminum bronzes are used for their combination of high strength, excellent corrosion and wear resistance. Aluminum bronze alloys typically contain 9-12% aluminum and up to 6% iron and nickel. Alloys in these composition limits are hardened by a combination of solid solution strengthening, cold work, and precipitation of an iron rich

phase. High aluminum alloys are quenched and tempered. Aluminum bronzes are used in marine hardware, shafts and pump and valve components for handling seawater, sour mine waters, and industrial process fluids. They are also used in applications such as heavy duty sleeve bearings, and machine tool ways.

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d. Silicon bronze: - it contains 1 to 4% Si, 0.25% to 1.25% Mn, 0.5% to 1% iron (if added) and the rest copper. Small addition of upto 0.5% Pb will improve machinability. It has high strength, toughness, and corrosion resistance. It can be readily hot worked. With low silicon content it can be safely cold worked also. It is widely used for boiler parts, tanks marine hardware and similar other item.

e. Manganese bronze: - It contains 55% to 60% copper, 38% to 42% Zn, upto 1.5% tin, upto 2% iron,

upto 1.5% Al and upto 3.5% Mn. It has superior mechanical properties and high corrosion resistance. It has poor respond to cold working but can be readily hot worked. It is used for such part

where high strength and corrosion resistance are desired, such as in ship propellers and rudders, etc.

Aluminum alloys

a. Duralumin: - It contains 4% Cu, 0.5% Mg, 0.5% Mn and the rest aluminum. Duralumin alloys are relatively soft, ductile, and workable in the normal state; they may be rolled, forged, extruded, or drawn into a variety of shapes and products. Their light weight and consequent high strength per unit weight compared to steel suit them for aircraft construction. Because duralumin loses strength in welding, a special laminated sheet form called alclad is used for aircraft construction; it has thin surface layers of pure aluminum or a corrosion-resistant aluminum alloy covering the strong duralumin core.

b. Aluminum casting alloy: - It contains 90% Al, 8% Cu, 1% iron and 1% Si. It has good strength, hardness and machinability. It may be sand, pressure or gravity die cast.

c. Y-alloy: - It contains 93% Al, 4% Cu, 2% Ni, and 1% Mg. Its principle use is as a casting alloy. It maintains its strength at elevated temperatures, and is used for pistons of I.C. engines. A treatment of Y-alloy castings, consisting of quenching in boiling water from a temperature of 5100 C and then aging for 5 days, develops very good mechanical properties in them. It is also used in strip and sheet forms.

Nickel alloys

a. German silver: - The name German silver or nickel silver is misleading since it has no silver content, but is an alloy of 60% Cu, 20% Ni and 20% Zn. It resembles silver but instead of the soft white color it has a yellow-grey tinge.

German silver is used extensively in the hollow-ware and flatware industries as E.P.N.S. (electro-plated nickel silver), which is a nickel silver item with a plating of silver. Nickel silver is slightly more difficult to work for jewelry purpose, than copper and other metals, since it is a harder alloy. Again it is prone to tarnish on exposure to air and will turn the skin green with prolonged contact.

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- b. Constantan: - It contains 45% Ni, 55% Cu. It has high specific resistance, which is unaffected by temperature variation. It is used for accurate resistors, thermocouples, Wheatstone bridge, low temperature heater and resistances.
- c. Monel metal: - It contains 68% Ni, 30% Cu, 1% iron and small additions of Mn and other elements. It is a nickel-copper alloy with high strength and excellent corrosion resistance in a range of media including sea water, hydrofluoric acid, sulfuric acid, and alkalis. Used for marine engineering, chemical and hydrocarbon processing equipment, valves, pumps, shafts, fittings, fasteners, and heat exchangers.
- d. Inconel: - It contains 80% Ni, 14% Cr and 6% iron. Inconel Alloys Are Generally Known For Their Resistance To Oxidation And Their Ability To Maintain Their Structural Integrity In High Temperature Atmospheres. There Are Several Inconel Alloys That Are Used In Applications That Require A Material That Does Not Easily Succumb To Caustic Corrosion, Corrosion Caused Purity Water, And Stress-Corrosion Cracking. While Each Variation Of Inconel Has Unique Traits that make it effective in different circumstances, the majority of the alloys are used frequently in the chemical industry.
- e. Nichrome: - Like Inconel it is also a nickel chromium alloy which is extensively used in electrical appliances as resistance wire.
- f. Incoloy: - It is also a Ni based alloy which is widely used as a high temperature alloy. It consists of 42% Ni, 13% Cr, 6% Mo, 2.4% Ti, 0.04% C and rest iron.
- g. K-Monel: - It possesses similar composition as monel, but about 3 to 4% Al is added to it. It carries similar application as monel, but has better mechanical properties than that.
- h. Nimonic alloy: - It contains 80% Ni and 20% Cr. It has high strength and ability to operate under intermittent heating and cooling conditions. It is widely used in gas turbine engines.

Bearing metals

Bearing metal should possess the following important characteristics:

1. It should have enough compressive strength to possess adequate load carrying capacity.
2. It should have good plasticity to allow for small variation in alignment and fitting.
3. It should have low co-efficient of friction to avoid excessive heating.

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Some important bearing metals are the following:

- a. Babbitt metal: - It is white metal containing 85% tin, 10% Sb and 5% Cu. It is used for heavy duty vehicles.
- b. Lead alloy: - It contains 40% lead and 60% Cu. It may be cast in position or fused as a thin shell to a bronze or steel reinforcing shell outside.
- c. Phosphor bronze: - With 10% tin is used for light load low speed bearings. It can be sand and centrifugal cast.

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Paper 4 of 4 — MAR 324 POLYMERS AND ELASTOMERS

Level 300 • Second Semester • Handout

ORGANIC MATERIALS

POLYMERS AND ELASTOMERS

1. INTRODUCTION

The materials, which are derivatives of carbon, chemically combined with hydrogen, oxygen or any other non-metallic substance, and their structures, in most cases, fairly complex are known as organic materials. Organic materials may be natural or synthetic, i.e. prepared or manufactured artificially. The naturally available organic materials include wood, natural rubber, coal, petroleum and food products, etc. The synthetic organic materials include, plastics, lubricants, rubber, soap oils, synthetic fibres, etc. Obviously, the range of organic compounds is very extensive as the thousands of hydrocarbon compounds and their derivatives are in existence. Plastics and synthetic rubbers termed polymers and wood and wood products are common engineering materials of industrial importance.

The organic materials are marked by their non-crystalline nature. However, the non-crystalline solid structures of organic materials is not completely random as they are composed of ordered atoms in several sub-units. The structures of all non-crystalline materials are such that the sub-unit arrangements are tangled very easily and completely in liquid state itself. However, on solidification it is impossible for them to get untangled. Such materials result into three dimensional network or long chain molecules due to the entanglement. These non-crystalline organic materials may be elements or compounds and are usually referred to as amorphous solids. Glass is a common example of these materials. SiO (silica) is another example of amorphous materials. The most important amorphous materials are polymers.

2. POLYMERS

As with metals and ceramics, the properties of polymers are intricately related to the structural elements of the material. Most polymers are organic in origin and are based on hydrocarbons, i.e. they are composed of hydrogen and carbon. Moreover, the intramolecular bonds are covalent.

Most polymers are organic (carbon-based) materials that contain molecules composed of various combinations of hydrogen, oxygen, nitrogen and carbon. These four elements are among the most common found in organic polymers. Carbon forms the “spine” of the polymer chain, and the other constituents attach themselves to the carbon. These polymer chains become entangled and form irregular coils which give them added

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strength. A portion of this entanglement is natural and can be further induced by additives and controlled processes. Most polymers are based on hydrocarbons, where the elements of carbon and hydrogen form predictable combinations based on the relationship, C_nH_{2n+2} . These petrochemical intermediates are chemicals manufactured from paraffins in petroleum and natural gas, which are further processed into polymer products.

General characteristics of polymers can be summarized as follows:

(i) Polymers have long chain structures. The individual molecule of a polymer is very large, i.e., it may consist of thousands of similar small molecules, all bonded together covalently.

(ii) All polymers have one thing common, i.e., carbon, which further bonds with hydrogen, nitrogen, halogens or other organic or inorganic substances.

(iii) Although, polymer's structure may be crystalline in simple materials but generally they are non crystalline solids at room temperatures. No doubt, polymers pass through a viscous stage during formation.

(iv) Polymers have light weight and they can be easily fabricated and shaped. They are poor conductors of electricity and their thermal conductivity is also low. Moreover, the polymers are resistant to chemical attack and decay.

Polymers are widely used in different fields. Polymers are used in indefinite number of forms, e.g. as telephone sets, paints, radio and television cabinets, coatings, adhesive and other countless objects. Few notable developments in polymer materials are:

(i) polymer materials far stronger than steel according to strength to weight ratio.

(ii) polymer materials suitable for repairing damaged kidney or heart.

(iii) materials act as adhesive so strong that the use of nails in wood and leather can be avoided.

(iv) lining for spacecraft to withstand high temperature, especially during re-entering into earth's atmosphere. We know that there is increase in temperature due to friction when the spacecraft re-enters the earth's atmosphere.

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(v) silicon film does not permit water to enter but allows dissolved oxygen, in water, to pass through it. Using polymers, under water tents have been prepared which allows a man to live and work.

3. BROAD CLASSIFICATION

Usually polymers are broadly classified in three categories:

(i) Plastics (ii) Fibres and (iii) Elastomers.

Also polymers are also classified according to their composition, formability,

geometrical structure and molecular structure.

Plastic:

These are the organic materials that can be easily shaped or moulded by mechanical or chemical action, with or without the application of heat.

Obviously, a plastic is a permanently deformable or mouldable material.

Probably, the largest number of different polymeric materials fall under this category. Polyethylene, polypropylene, polyvinyl chloride, polystyrene, and fluorocarbons, epoxies, phenolics and polyesters may be put under this category.

Plastics have a wide variety of combination of properties.

There are some plastics which are very rigid and brittle, whereas others are flexible, exhibiting both elastic and plastic deformations when stressed, and some time experiencing sufficient deformation prior to fracture.

Polymers belonging to this class of materials may have any degree of crystallinity, and all molecular structures and configurations (i.e., linear, branched, isotactic, etc.) are possible. Plastic materials may further be classified as thermoplastic or thermosetting. Table below summarizes the characteristics and typical applications for a number of plastics. Plastic have following properties:

- (i) Plastics are light weight materials that is their densities and specific gravities are low.
- (ii) They have high resistance to chemical attack, i.e. corrosion.
- (iii) They have very good surface finish when obtained from the dies.
- (iv) They have high thermal and electrical insulation.
- (v) As compared to metals their strength is low and also low elastic modulus value.
- (vi) As compared to metals, plastic have low softening and thermal degradation temperature.
- (vii) Plastics are poor to fair dimensional stability, particularly in moist conditions.
- (viii) Plastics are available in different colours as well as transparent form.
- (ix) With respect to metals, plastics have high coefficient of thermal expansion.
- (x) These are unsuitable for high temperature applications.
- (xi) Plastics can be easily fabricated.

Plastic Trade Characteristics Typical
material (Type) names applications

Thermoplastics

Acrylonitrilebutadiene- Abson Outstanding Refrigerator
styrene Cylolac strength and linings, lawn

(ABS Kralastic toughness, and garden
 Lustran resistant to heat equipment,
 Novodur distortion; good toys, high-way
 Tybrene electrical safety
 properties; devices
 flammable
 and soluble in
 some organic
 solvents
 Acrylics (polymethyl Acrylite Outstanding Lenses,
 methacrylate) Diakon light transparent
 Lucite transmission aircraft
 Plexiglas and enclosures,
 resistance to drafting
 weathering; equipment,
 only fair outdoor
 mechanical signs
 properties

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Fluorocarbons Teflon Chemically Anticorrosive
 (PTFE or TFE) Fluon inert in almost seals, chemical
 Halar all Pipes and
 Halon environments, valves,
 Hostaflon TF Excellent bearings,
 electrical antiadhesive
 properties; coatings, high-
 low coefficient temperature
 of friction; electronic
 may be used to parts
 260°C (500°F);
 relatively weak
 and poor cold-
 flow properties
 Polyamides Nylon Good Bearings,
 (nylons) Durethan mechanical gears, cams,
 Herox strength, bushings,
 Nomex abrasion handles, and
 Ultramid resistance, and jacketing for
 Zytel toughness; low wires and
 coefficient of cables
 friction; absorbs

water and some
 other liquids
 Polycarbonates Baylon Dimensionally Safety
 Iupilon stable; low helmets,
 Lexan water lenses, light
 Makrolon absorption; globes, base
 Merlon transparent; for
 Nuclon very good photographic
 impact film
 resistance and
 ductility;
 chemical
 resistance not
 outstanding
 Polyethylene Alathon Chemically Flexible
 Alkathene resistant and bottles, toys,
 Ethron electrically tumblers,
 Fortiflex insulating; B attery parts,
 Hi-fax tough and ice trays, film
 Petrothene relatively low wrapping
 Rigidex materials
 5
 Zendel coefficient of
 friction; low
 strength and
 poor resistance
 to weathering
 Polypropylene Bexphane Resistant to heat Sterilizable
 Herculon distortion; bottles,
 Meraklon excellent packaging
 Moplen electrical film, TV
 Poly-pro properties and cabinets,
 Pro-fax fatigue luggage
 Propathene strength;
 chemically
 inert;
 relatively
 inexpensive;
 poor resistance
 to UV light
 Polystyrene Carinex Excellent Wall tile,

Celatron electrical battery cases,
 Hostyren properties and toys, indoor
 Lustrex optical clarity; Lighting
 Styron good thermal panels,
 Vestyron and appliance
 dimensional housings
 stability;
 relatively
 inexpensive
 Vinyls Darvic Good low-cost, Floor
 Exon general-purpose coverings,
 Geon materials; pipe, electrical
 Pee Vee Cee ordinarily rigid, wire
 Pliovic but maybe insulation,
 Saran made flexible garden hose,
 Tygon with phonograph
 plasticizers; records
 often
 copolymerized;
 susceptible
 to heat
 distortion

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Polyester (PET Celanar One of the Magnetic
 or PETE) Crastin toughest of recording
 Dacron plastic films; tapes, clothing,
 Hylar excellent automotive
 Melinex fatigue and tear tire cords,
 Mylar strength, and beverage
 Terylem resistance to containers
 humidity, acids,
 greases, oils,
 and solvents
 Thermosetting Polymers
 Epoxies Araldite Excellent Electrical
 Epikote combination of moldings,
 Epon mechanical sinks,
 Epi-rez properties and adhesives,
 Lekutherm corrosion protective
 Nepoxide resistance; coatings, used
 dimensionally with fiberglass

stable; good laminates

adhesion;

relatively

inexpensive;

good

electrical

properties

Phenolics Bakelite Excellent Motor

Amberol thermal stability housings,

Aroclene to over 150°C telephones,

Durite (300°F); may be auto

Resinox compounded distributors,

with a large electrical

number of fixtures

resins, fillers,

etc.;

inexpensive

Polyesters Aroclene Excellent Helmets,

Baygal electrical fiberglass

Derakane properties and boats, auto

Laguval low cost; can be body

Laminac formulated for components,

Selectron room- or high- chairs, fans

temperature

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use; often fiber

reinforced

There are many plastics with outstanding properties. Polystyrene and polymethyl methacrylate are especially useful for applications in which optical transparency is critical, however, it is imperative that the material be highly amorphous or, if semicrystalline, have very small crystallites. The fluorocarbons have a very low coefficient of friction and moreover they are extremely resistant to attack by a host of chemicals, even at relatively high temperatures. They find wide range of uses as coatings. On nonstick cookware, in bearings and bushings, and for high temperature electronic components.

Thermoplastic resins are generally referred to as plastics and have the property of increasing plasticity, i.e., ability to deform plastically with rise in temperature. They have the long chain structure. Unlike thermosetting resins they not get soft. Thermoset polymers are generally harder and stronger than thermoplastics and have better dimensional stability.

Thermoplastic polymers include acrylic, nylon (polyamide), cellulose,

polystyrene, polyethylene, fluorocarbons, and vinyl. These are used for plastic walls and floor tiles, fluorescent light, plastic lenses, etc. Common thermosetting polymers include phenolics, amines, polyesters, epoxies and alkyds

Major characteristics of plastics are:

- (i) Low softening temperature
- (ii) Non-crystalline structure
- (iii) can be readily formed into desired complex shapes
- (iv) With some exceptions resistant to chemical reaction
- (v) poor conductor of electricity and low thermal conductivity and
- (vi) visco-elastic behaviour.

Elastomers:

These are also long chain molecules, but exhibit the unique property of high elasticity. These materials have properties similar to polymers. These materials may be repeatedly stretched or elongated and will return to their original condition upon release of the force producing the elongation. They exhibit elastic behaviour, as compared to polymers, which exhibit greater plastic properties. Included in this category are rubber and rubber like materials. A distinction is made between rubber and elastomers. Rubber must withstand a 200% elongation and rapidly return to its original dimensions. This property of a material to recover from elastic deformation is termed as resilience

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Important characteristics and typical applications of few common elastomers

Material Trade/ Elongation(Useful Characteristics Application

(Chemical composition %) temperatures (typical)

Type) name range

(deg)

Natural Rubber 500-760 -60 to 120 Excellent Pneumatic

poly- rubber physical tyres and

isoprene (NR) properties; tubes;

good heels and

resistance to soles;

cutting, gaskets

gouging, and

abrasion;

low heat,

ozone,

and oil

resistance;

good

electrical

properties

Styrene- GRS, 450-500 -60 to 120 Good Same as

butadiene Buna S physical natural

copolymer (SBR) properties; rubber

excellent

abrasion

resistance;

not oil,

ozone,

or weather

resistant;

electrical

properties

good,

but not

outstanding

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Acrylonitril Buna A, 400-600 -50 to 150 Excellent Gasoline,

e- Nitrile resistance to chemical,

butadiene (NBR) vegetable, and

copolymer animal, and oil hose;

petroleum seals and

oils; poor Orings;

lowtemperatu heels and

re soles

properties;

electrical

properties not

outstanding

Chloropren Neopre 100-800 -50 to 105 Excellent Wire and

e ne (CR) ozone, heat, cable;

and chem.

weathering tank

resistance; linings;

good oil belts,

resistance; hoses,

excellent seals, and

flame gaskets

resistance;

not as good in

electrical

applications

as

natural

rubber

Polysiloxan Silicone 100-800 -115 to Excellent High-and

e (VMQ) 315 resistance to low-

high and low temperatur

temperatures; e

low strength; insulation;

excellent seals,

electrical diaphragm

properties s;

tubing for

food

and

medical

uses

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The properties of common elastomers are typical and, of course, depend on the degree of vulcanization and on whether any reinforcement used. Natural rubber has an outstanding combination of desirable properties and still utilized to a large degree. However, the most important synthetic elastomer is SBR (styrene butadienecopolymer) which is used especially in automobile tyres, reinforced with carbon black. Acrylonitrile butadiene copolymer (NBR) is another common synthetic elastomer, which is highly resistant to degradation and swelling.

Fibres:

These consists of long chain molecules with all the chains aligned to more or less longitudinal direction of the fibre. The alignment shows highly directional behaviour with strength along the length being more than ten times that across the length. The directional properties are improved by alignment, i.e. the strength of fibres are greatly enhanced in the direction parallel to the fibre length. These provide strength to the molecular units.

The fibre polymers are quite capable of being drawn into long filaments having at least a 100 : 1 length to diameter ratio. Some polymers used as fibres, such as nylon and cellulose acetate, serve equally well as plastics. Most commercial fibre polymers are utilized in the textile industry, being woven or knit into cloth or fabric. In addition, the aramid fibres are widely employed in composite materials. A fibre polymer, to be used as a textile material must have some restrictive physical and chemical properties. Fibres, while its use may be subjected to a variety of mechanical deformations, e.g. stretching, twisting, shearing and abrasion. Obviously, they must have a high tensile strength over a

relatively wide temperature range and a high modulus of elasticity and also abrasion resistances. We may note that these properties are governed by the chemistry of polymer chains and also by fibre drawing process.

Synthetic fibres are produced by the following methods: (i) Melt spinning (ii) Dry spinning and wet spinning. To improve fibres properties, i.e., to increase/decrease resistance, water or fire proofs, etc., chemical treatment is done.

Metallic and ceramic fibres have also similar qualities as that of polymeric fibres. However, metallic and ceramic fibres have very high strength and therefore they have many industrial applications. There are also natural fibres, e.g. cotton, silk, wool, cellulose, etc. Cellulose fibres are flexible and are strong in tension. Cotton is produced naturally as a plant fibre. However, the synthetic fibres are cheap, durable and have more dimensional stability and that is why they have almost replaced the natural fibres.

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4 MISCELLANEOUS APPLICATION OF POLYMERS

Adhesives

Polymers adhesives are widely used to join a large variety of material combinations; e.g. metal-metal, metalplastic, metal-ceramic, etc. The service temperature limitation is a major primary drawback. The strength of organic polymers decreases rapidly with rise in temperature but they maintain their mechanical integrity only at relatively low temperatures.

Films

Polymeric materials have recently found widespread use in the form of thin films. Films, whose thicknesses lies between 0.025 and 0.125 mm are fabricated and widely used as bags for packaging food products and other items, as textile products, and a host of other uses. The main characteristics of the materials produced and used as films include low density, a high degree of flexibility, high tensile and tear strengths, resistance to attack by moisture and other chemicals, and low permeability to some gases, especially water vapour. The polymers which satisfy these criteria and are manufactured in film form include polyethylene, polypropylene, cellophane, and cellulose acetate, etc.

Foams

These are the plastic materials which contain a relatively high volume percentage of small pores. Thermoplastic as well as thermosetting materials are used as foams. These materials include polyurethane, rubber, polystyrene, and polyvinyl chloride. Foams are widely used as cushions in automobiles and furniture as well as in packaging and thermal insulations. The process of foaming is carried out by incorporating into the batch of material a blowing agent that upon heating, decomposes with the liberation of a gas. Gas bubbles are created throughout the now fluid-mass, which remains as pores upon cooling and yield

to a sponge-like structure. One can obtain the same effect by bubbling an inert gas through a material while it is in molten state.

4 POLYMER ADDITIVES

The properties of the polymers discussed so far are intrinsic ones, i.e., characteristic of or fundamental of or fundamental to the specific polymer. Many times, one finds it necessary to modify the mechanical, chemical, and physical properties to a greater degree than is possible by the simple alteration of this fundamental molecular structure. To enhance or modify many of these properties, foreign substances called additives are knowingly introduced so that a polymer material may be found more serviceable. Following are the typical additives:

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(i) Fillers: Many plastics are virtually useless alone but are converted into highly serviceable products by combining them with particulate or fibrous solids. This improves their tensile and compressive strengths, abrasion resistance, toughness, dimensional and thermal stability, and other properties. The principal fillers used can be divided into two types: particulate and fibrous. Among the particulate fillers are silica products, including sand, quartz and diatomaceous earth; silicates, including clay, mica, talc, asbestos, etc. Fibrous fillers, some quite old (cellulosics) and some new "space-age" products (metallic fibres, whiskers) include cellulosic fibres, such as alphacellulose and cotton flock

In recent years the use of mineral fillers has increased markedly, with emphasis on calcium carbonate. As these inexpensive materials replace some volume of the more expensive polymer, the cost of final product is reduced.

(ii) Plasticizers: These are added to plastic materials to improve flow and, therefore, processability, and to reduce the brittleness of the product. Thus the addition of plasticizers to plastics improves their flexibility, ductility, and toughness of polymers. Moreover, their presence also produces reductions in hardness and stiffness. Usually, plasticizers are liquids having low vapour pressures and low molecular weights. The small plasticizer molecules occupy positions between the large polymer chains thereby effectively increasing the interchain distance with a reduction in the secondary intermolecular bonding. Plasticizers are commonly used in polymers to make them soft, flexible and tough materials from those of a hard, brittle, glass like solid. An example is the plasticization of poly (vinyl chloride) and vinylchloride acetate copolymers

(iii) Stabilizers: Under normal environmental conditions, some polymeric materials are subject to rapid deterioration, usually in terms of mechanical integrity. Usually, this deterioration is a result of exposure to light, in particular ultraviolet (UV) radiation, and also oxidation. UV radiation interacts with and causes a severance of some of the covalent bonds along the molecular chain, which may also result in some crosslinking. Oxidation deterioration is a consequence of the

chemical interaction between oxygen atoms and the polymer molecules. Additives that counteract these deteriorative processes are named as stabilizers. An important example is vinyl resins. Other applications of stabilizers include the use of carbon black to prevent photochemical degradation (by excluding light), as in polyethylene, and of ultraviolet light absorbers, such as hydrobenzophenones, to improve the light stability of both plastics and their colorants.

(iv) Colorants: Colorants for plastics include a wide variety of inorganic and organic materials. These impart a specific colour to a polymer. Colorants may be added in the form of dyes or pigments.

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(vi) Flame Retardants: The flammability is a major concern for polymeric materials, especially in the manufacturing of textile goods and children's toys. In pure form, most of the polymers are flammable; exceptions are those polymers which contain significant contents of chlorine and/or fluorine, such as polyvinyl chloride and polytetrafluoroethylene. One can enhance the flammability resistance of the remaining combustible polymers by additives called flame retardants. These flame retardants may function by initiating a chemical reaction that causes a cooling of the combustion region and a cessation of burning. The most useful material imparting flame retardance to plastics is antimony trioxide.

5 REINFORCED POLYMERS

These are polymers whose properties have been altered through the use of reinforcement material. Glass, mineral and carbon fibres can be added to polymers to alter their properties. Most polymers are available as glass fibre-reinforced products. The glass fibres, whose diameters range from 0.0002 to 0.001 inch in diameter, are coated with a resin and a coupling agent. Property increase of 200% or better can be obtained through the glass fibre reinforcement.

The strength of these reinforced polymers depends on the type and quantity of filler used and the degree to which the resin has wetted the reinforcement material. One of the more common filler materials is glass fibre.

Carbon fibre-reinforced materials are more expensive than fibre-reinforced materials. Carbon fibre-reinforced materials have high tensile strength, stiffness, and greater related mechanical properties. They also produce lower coefficients of expansion, improved creep resistance, better wear resistance, greater toughness, and higher strength-to-weight ratios than glass fibre-reinforced materials.

Glass and carbon are the most common fibre reinforcements. Other materials used in fibre reinforcements include graphite, boron, and cotton.

Reinforcement materials are available in several forms, including mat, fibre bundles, chopper fibres, and cloth.

Most of the reinforced thermosetting polymers (90% to 95%) are polyesters and epoxies, with the majority being polyesters.

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These reinforced polymers are manufactured through a variety of methods, including hand lay-up, spray-up, matched moulding, premixed moulding, and vacuum-or pressure bag casting.

These materials are replacing metals in a variety of applications. They offer similar or better strength, characteristics at lower weight and often, at lower cost. They are frequently easier to produce, are resistant to corrosion and chemicals, and offer a wide variety of manufacturing processes, from the very simple and inexpensive to the higher volume and more intricate processes.

6 MECHANICAL PROPERTIES OF POLYMER

It is observed that the mechanical strength of polymers increases with the degree of crystallinity. This shows that the mechanical strength of polymer materials is a sensitive function of the process variables in the synthesis of these materials and therefore for the reproducibility and reliability of the products they should be properly controlled.

The increase of the strength of the polymer with crystallinity of the polymer

The properties of polymers, besides the degree of crystallinity, also depend on several parameters, e.g. nature of monomer unit, molecular weight, cross-linking, etc. This is why, it is very difficult to correlate the mechanical properties of the polymers with molecular structure directly. When subjected to external stresses, many polymers undergo not only elastic deformation but also viscous flow.

There are some textile fibres which show a linear relationship between stress and strain, i.e. stress strain, for moderate stresses. However, beyond a

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certain point, the elongation increases rapidly. We may note that polycrystalline structure does not change on the application of stress, e.g., in the case of nylon, such a relationship is almost entirely linear.

Concluding, we can say that in polymers, molecular structure which is controlled by synthesis parameters influences the properties to a great extent.

7 FABRICATION OF FIBRES AND FILMS

Fibres: The process by which fibres are formed from bulk polymer material is called spinning. Most often, fibres are spun from the molten state in a process called melt spinning. First, the material to be spun is heated until it forms a relatively viscous liquid and then it is pumped down through a plate called a

spinnerette, which contains numerous small round holes. When the molten material passes through each of these orifices, a single fibre is formed, which solidifies almost immediately upon passing through the air.

The crystallinity of a spun fibre will depend on its rate of cooling during the process of spinning. The strength of fibres is improved by a postforming process called drawing. However, drawing is simply the mechanical elongation of a fibre in the direction of its axis. The molecular chains during this process become oriented in the direction of drawing, such that the tensile strength, modulus of elasticity, and toughness are improved to a certain extent. Although the mechanical strength of a drawn fibre is improved in this axial direction, strength is reduced in a transverse or radial direction. Moreover, since fibres are normally stressed only along the axis, this strength differential is not critical. We may note that the crosssection of drawn fibres is nearly circular, and the properties are uniform throughout the cross-section.

Films: Many films are simply extruded through a thin die slit and then this may be followed by a rolling operation that serves to reduce thickness and improve strength. Alternatively, film may be blown: continuous tubing is extruded through an annular die; then, by maintaining a very carefully controlled positive gas pressure inside the tube, one may reduce wall thickness continuously to produce a thin cylindrical film, which may be cut and laid flat. Some of the newer films are produced by coextrusion; that is, multilayers of more than one polymer type are extruded simultaneously.

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